



GRAND CANYON
UNIVERSITY™

Pi-Piper Project Requirements

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CST-451 Capstone Project Requirements Document

Grand Canyon University

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ABSTRACT

Republic Pi's current order management system involves manually comparing prices between Sysco and US Foods using a paper prep list and an iPad. To streamline this process, the proposed app will fully digitize and automate the restaurant's prep list and order management system. On the admin side of the app it will compare prices between suppliers, generate the most cost-effective orders, and offer suggestions based on past data and current needs. Also exclusively on the admin side will feature a prep list and ingredient CRUD tool. Additionally, the app will feature a visualized prep list and planning tool to assist employees in pre-open and pre-rush preparation. With presets and automated suggestions, the app reduces the need for manual input, optimizing inventory restocking and prep processes. The aim is to create a lightweight, dynamic admin system that saves time, increases productivity, and requires minimal oversight, ensuring that managers can focus on other critical aspects of running the restaurant.

History and Signoff Sheet

Change Record

Date	Author	Revision Notes
10/10	P.Hollenback	Initial draft for review/discussion

Overall Instructor Feedback/Comments

Overall Instructor Feedback/Comments

Integrated Instructor Feedback into Project Documentation

☐ Yes ☐ No

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Functional Requirements

Use Cases

View the attached User stories to this submission.

Non-Functional Requirements

Use Cases(NFR):

My NFR is **Data backup**. Because there is major emphasis upon the data driven components of my application, such as BI reports as detailed below, or daily suggestions defined by inventory management and ingredient pricing, ensuring that there are no holes in time-series data will ensure that my data is up to date and provides useful, accurate insights.

The backup redundancy I am aiming for is 24 hours. At worst there can be one day's worth of data loss. My decision of 24 hours is due to the fact that many decisions being made that go into data can be recalled in the case of a loss of a day of data. One day is about the biggest amount that would be possible to manually fill in, any more may induce some issues by user entry error.

The Recovery Point Objective (RPO) that I would like to aim for is sometime in the early morning, somewhere between 2am and 5am so that there is zero conflict with the restaurant operations. Sometimes there can be planning of prep or managerial duties that run into the early morning if there was a late close the day before but I figure there will never be employees or managers in at these hours no matter the case. The key point that I want to hit with backing up data between these times is that the overhead of the application does not interfere with backing up, and flags thrown by a bad data backup can be signaled to management so it can be fixed first thing in the morning.

I plan to achieve this by purchasing a hard disk connected to my in house hosting machine. There are numerous advantages that can be linked to utilizing a hard disk. First is the accessibility, because hard disks generally utilize a wired connection, there is little overhead for backing up data since they can be accessed so quickly. It is also relatively safe disaster wise especially in my case where it is right next to the hosting machine, in case of disaster there is less chance that the data gets corrupted in the hard disk rather than the computer which is more susceptible to water, or any other potentially harmful material.

Integrity of each backup should be checked upon backup completion, flagging any errors if data is incomplete or does not represent a full day's worth of data. Corrupted backups would be one of the most detrimental issues to plague my backup system, becoming unaware of poor backups would create problems if it had to be used to recover data and produce poor business insights.

Access control is an important aspect of this data backup system. Data backups and recoveries must only be initiated by authorized personnel. Failure to adhere by this principle could cause issues especially when trying to initiate a data recovery when not necessary, improperly overwriting data and likely creating errors. This will also ensure that when the data is backed up, it is out of the restaurant's hours of prep/service so that the performance overhead is focused on in-hours tools that may be used rather than backup services.

Putting all of these elements into practice will ensure consistent data storage and will limit the amount of data loss if something unforeseen were to happen. The documentation of these policies are also crucial because in order to maintain consistent practices, whether or not I am maintaining the system by first outlining the importance of these issues, the policies should come naturally from this

Technical Requirements

Use Cases

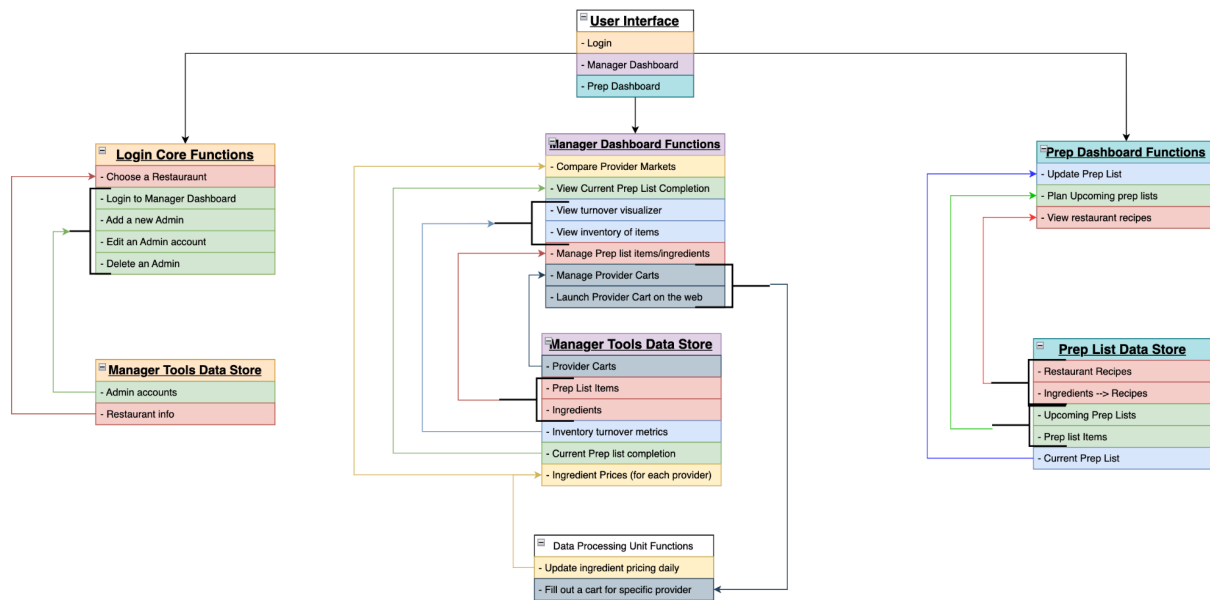
The following describe the tools and technologies used in this project:

Technology or Tool	Version	Justification
React	18	Frontend framework that will handle the state of all pages on the frontend to manage lists and orders before posting them to the server.
TypeScript	5.5.4	Ensuring strong typing will make sure the data is always consistent.
PWA	N/A	Creating an app where tools are available even when the connection may be poor.
Tailwind CSS	3.x	Offers lots of custom CSS in order to keep the CSS minimal and speed up development with utility classes.
PostgreSQL	14.x	Offers plenty of BI tools as well as visualization of time-series data, which is helpful in this context.
JWT	8.5.1	Secure authentication process for managers to gain access to the manager tools.
BeautifulSoup	4.10.0	Parses HTML to extract data from websites.
Selenium	4.0.0	Allows automation of sites to navigate a site and handle dynamically loaded content.
***	Still Under Review	***
Express.JS	18.x(LTS)	Writes JS on the server side of the application
Django	4.x	Keeping the data processing and web server in the same project would prove beneficial

Tableau	N/A	Tableau can connect directly with SQL databases to lessen the need for some business logic.
Chart.JS	2.x	Allows me to implement my own analytical logic, and create customizable reports.

Logical System Design

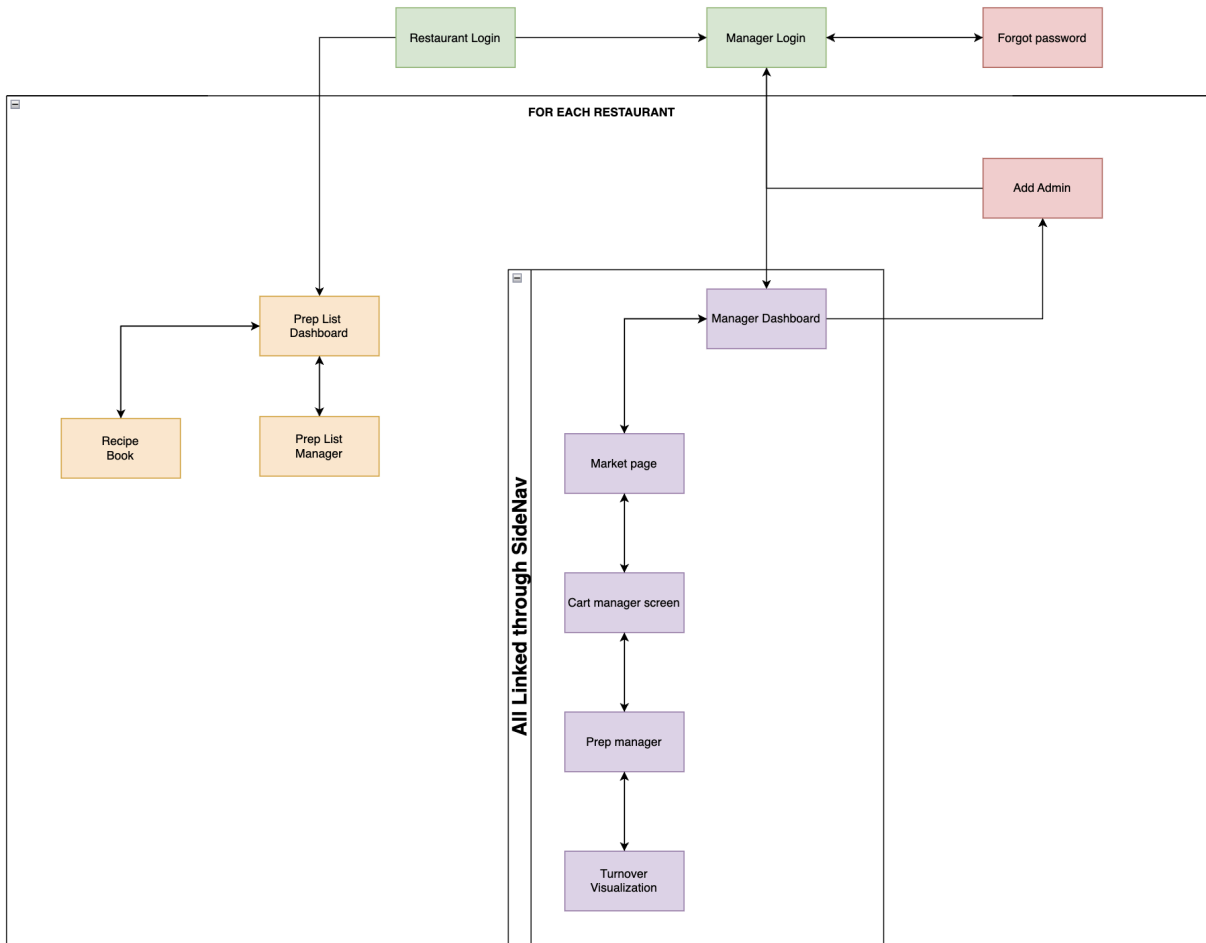
A diagram of the logical architecture of the system.



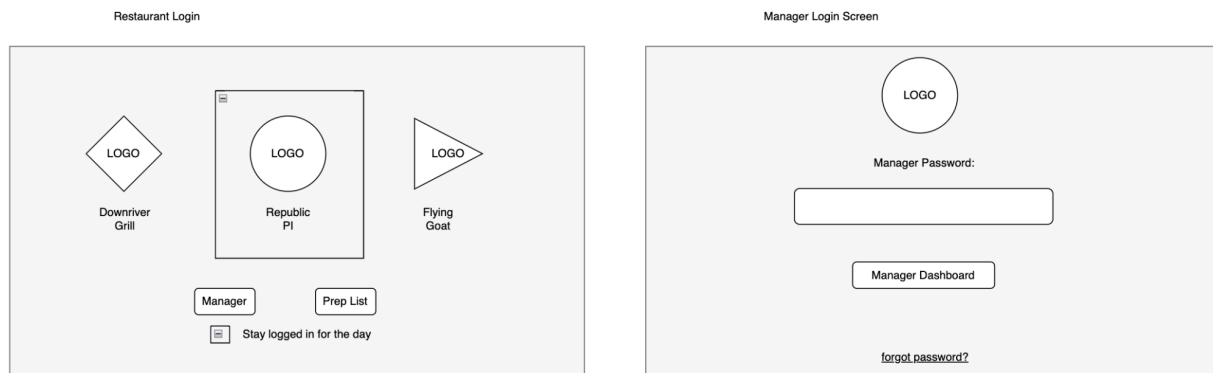
User Interface Design

A sitemap and user interface design diagram for each user interface screen in the application

Sitemap:



Login:



Prep List dashboard

Manager

search:

Prep Plan

Recipebook

Categories

Slicing

Pantry

Grill

Grill

To-Do

Completed

Slice Ham
4 loins

- Use Ham from walk-in
- Set slicer width to just over an inch
- Hold in pan and keep in walk-in

Slice Prosciutto
2 loins

- Use Prosciutto from cooler
- Set slicer width to just under an inch
- Wrap in saran and keep in walk-in

Slice Coppa
4 loins

- Use Coppa from walk-in
- Set slicer width to just under an inch
- Hold in pan and keep in walk-in

Slice Provalone
4 loins

- Use provalone from walk-in
- Set slicer width to just under an inch
- Hold in pan and keep in walk-in

Italian Sausage Mix
1 serv./ 24 oz

- Mix in machine
- Chill in the walk-in

Pi Bites
40

- Put in Oven for 40 minutes
- Slice 40 bags worth
- Store above grill

Prep Plan

Create Prep List For: Monday, 09/24

search:

Prep Dashboard

Slice Ham

>

Slice Prov.

>

Slice Pros.

>

Slicing

Cucumber

>

Romaine

>

Tomatoes

>

Pantry

Slice Ham
qty
slice the ham with the slicer. chill in the walkin until it reaches tempature

Slice Ham
qty
slice the ham with the slicer. chill in the walkin until it reaches tempature

Slice Ham
qty
slice the ham with the slicer. chill in the walkin until it reaches tempature

Recipe Book
search:

Prep Dashboard

recipe1

recipe2

recipe3

recipe4

recipe5

recipe6

recipe7

Recipe Name

1. Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.
2. Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.
3. Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.
4. Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor

Navigates To:

1. Market Page
2. Cart Manager
3. Turnover Visualizer
4. Prep manager

Manager Dashboard

Login

Add Admin

Prep

Time to Open:

2 hours
30 minutes

Local Time:

8:30 am

Inventory

Critical	- Pepperoni - Brussel Sprouts - Jalepano
Close	- Pineapple - Garbanzo Beans - Cheese Blend
Good	Rest of Inventory

Pantry

Phase 1 Phase 2 Phase 3

Oven

Phase 1 Phase 2 Phase 3

Grill

Phase 1 Phase 2 Phase 3

Market page

Navigates To:
ALL MANGER TOOLS

Cluster Popup

Create Item Group

Provider Market

Today's Suggestions:

Cucumbers			
last ordered:	12/23		
Sysco:	Us Foods:	QTY	
7.00 \$	7.50 \$	3	
☐ select	☐ select	A	D

Peccerino			
last ordered:	12/23		
Sysco:	Us Foods:	QTY	
7.00 \$	7.50 \$	3	
☐ select	☐ select	A	D

Olives			
last ordered:	12/23		
Sysco:	Us Foods:	QTY	
7.00 \$	7.50 \$	3	
☐ select	☐ select	A	D

Searchbar

Group Selector

Category Selector

Pineapple			
last ordered:	12/01		
Sysco:	Us Foods:	QTY	
7.10 \$	6.50 \$	0	
☐ select	☐ select	A	D

Pineapple			
last ordered:	11/25		
Sysco:	Us Foods:	QTY	
2.12 \$	2.18 \$	0	
☐ select	☐ select	A	D

Beets			
last ordered:	12/14		
Sysco:	Us Foods:	QTY	
3.00 \$	4.50 \$	0	
☐ select	☐ select	A	D

Eggs			
last ordered:	12/04		
Sysco:	Us Foods:	QTY	
8.00 \$	7.90 \$	0	
☐ select	☐ select	A	D

Mushrooms			
last ordered:	12/20		
Sysco:	Us Foods:	QTY	
2.00 \$	1.50 \$	0	
☐ select	☐ select	A	D

Brussel Sprouts			
last ordered:	12/18		
Sysco:	Us Foods:	QTY	
3.00 \$	4.50 \$	0	
☐ select	☐ select	A	D

Create a item grouping

grouping name:

☐ ingredient name
 ☐ ingredient name

☐ ingredient name
 ☐ ingredient name

☐ ingredient name
 ☐ ingredient name

☐ ingredient name
 ☐ ingredient name

Create

Prep Manager

Navigates To:
ALL MANGER TOOLS

Start managing:

Prep List Items

Ingredients

SearchBar

Category Selector

id	Title	Notes	Category	
1	Value 1	description	category	Edit
2	Value 1	description	category	Edit
3	Value 3	description	category	Edit
4	Value 4	description	category	Edit
5	Value 5	description	category	Edit
6	Value 6	description	category	Edit

ADD

DELETE

Navigates To:
ALL MANGER TOOLS

Cart Manager

Manage My Carts

sysco

ingredient name	qty +/-	price per unit	pricing status	Delete	Switch Provider

TOTAL:

X.XX

Open Order

US foods

ingredient name	qty +/-	price per unit	pricing status	Delete	Switch Provider

TOTAL:

X.XX

Open Order

Admin Management: (ADD)

name:

number:

password:

Create
(Back To Login)

VIEW ADMIN

ADMIN DELETE POPUP

◀ CLOSE

VIEW ADMIN

Current Admins

	Admin Name	EDIT	DELETE
1	Peyton B. Hollenback	EDIT	DELETE
2	Earl R. Monroe	EDIT	DELETE
3	Hunter S. Thompson	EDIT	DELETE
4	Richard L. Sherman	EDIT	DELETE
5	Bo M. Nix	EDIT	DELETE
6	Brian M. Thomas	EDIT	DELETE

UPDATE

delete admin?

DELETE

REVERT

admin deleted.

CLOSE

update success

CLOSE



forgot password?

***send a recovery password to
a registered admin:***

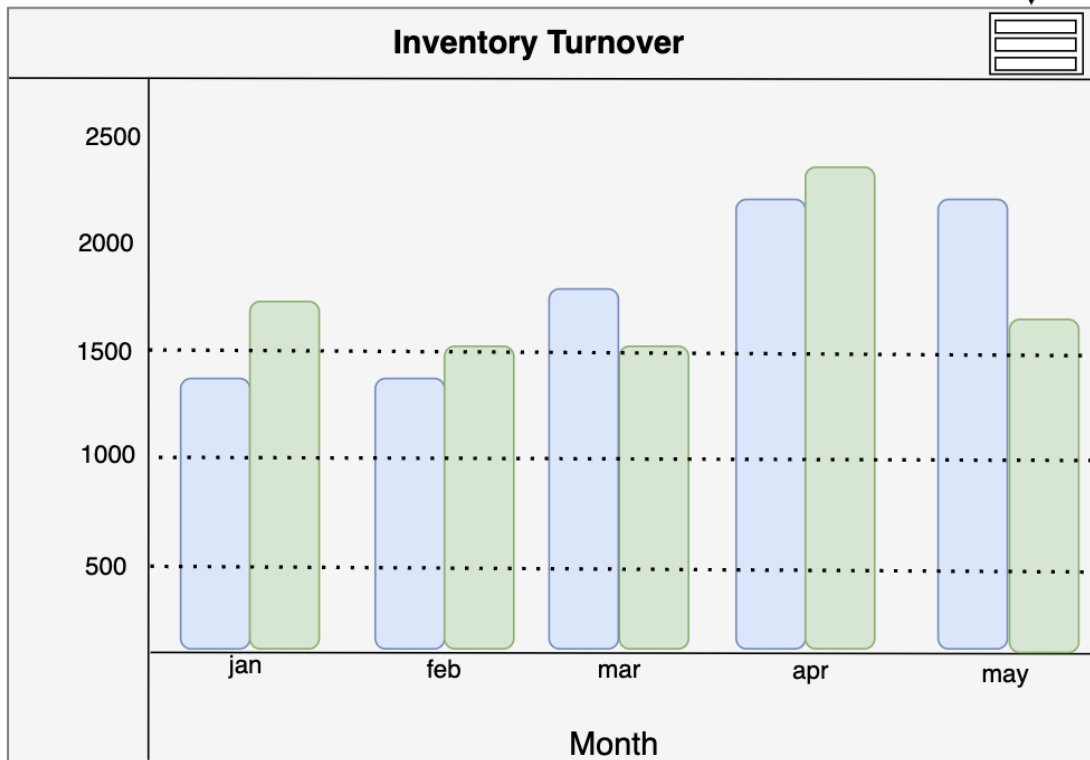
number:

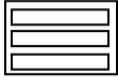


***reset password
sent.***

Turnover visualizer

Navigates To:
ALL MANGER TOOLS





This icon represents a side nav bar.
This will bring up this side screen:

Navigation
page 1
page 2
page 3
so on...

Reports Design

Inventory Turnover Report:

*** VIEW THE REPORT VIEW IN THE SECTION ABOVE ***

Purpose:

To track the turnover rate of inventory based on completed prep list and inventory ordering. This creates a metric that can be compared between multiple restaurants and can easily visualize performance in relation to what is expected from a location and what the result is on a monthly basis. This can create an opportunity for management to create incentive to be accurate with inventory turnover and complete orders that limit waste and don't sit on shelves for too long.

Data Included:

- Prep List Completion data
- Total Quantity Expected
- Total Quantity Completed
- Date of inventory Order

Formatting Requirements:**Bar Chart:**

Display total quantity requested VS Total Quantity completed each month the following are some requirements of the bar chart:

- > Aggregate Monthly Completion: display a single bar chart for all items monthly which can be broken down into specific items upon filtering.